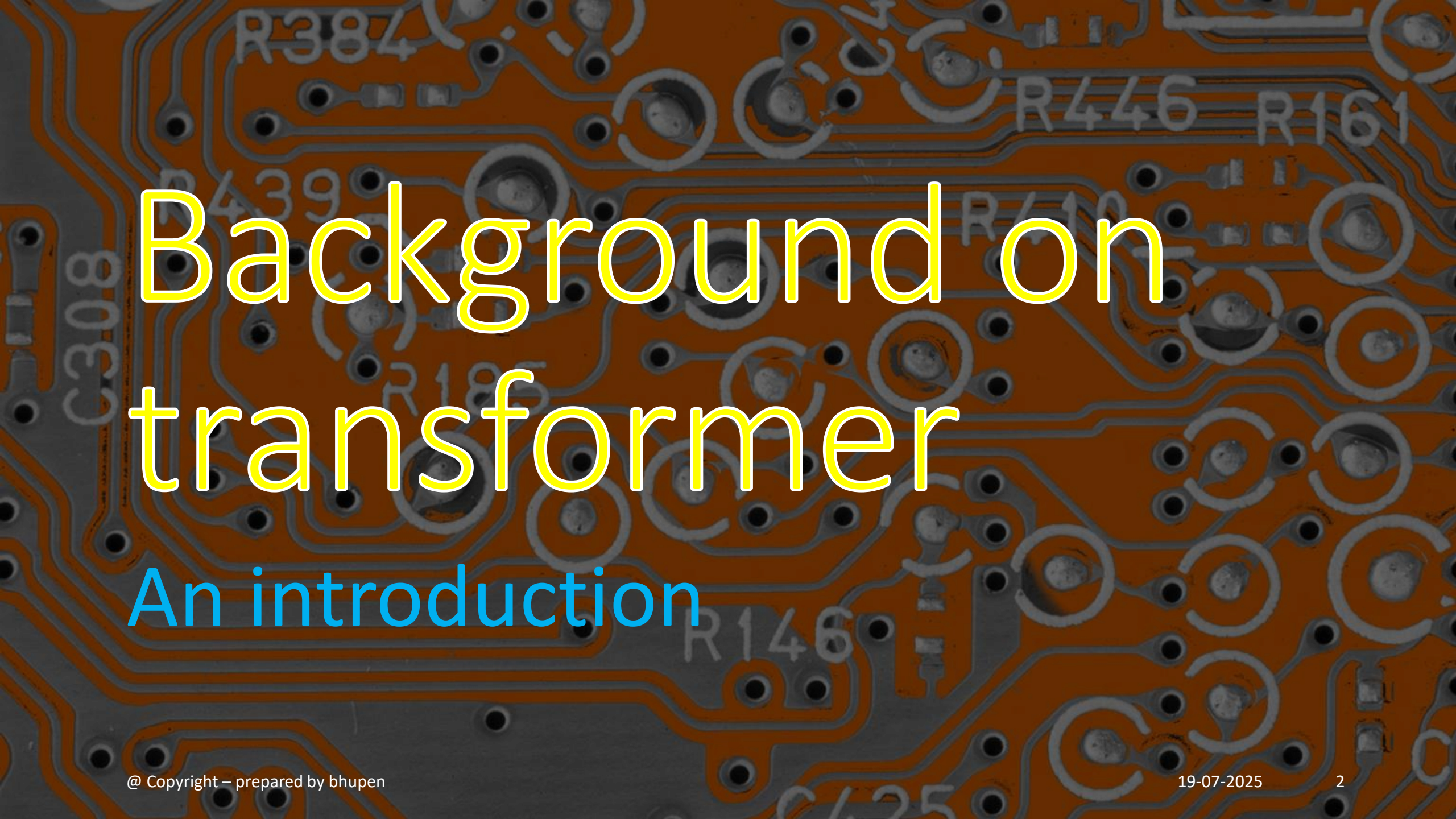


Transformer Networks

highly effective in various
NLP tasks and beyond



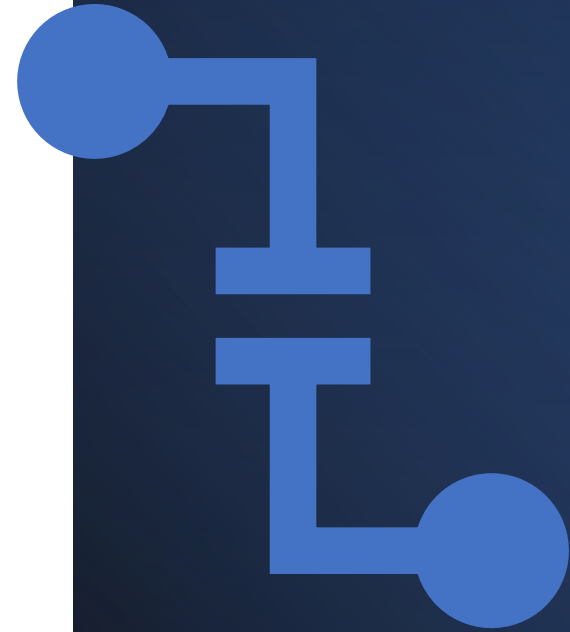


Background on transformer

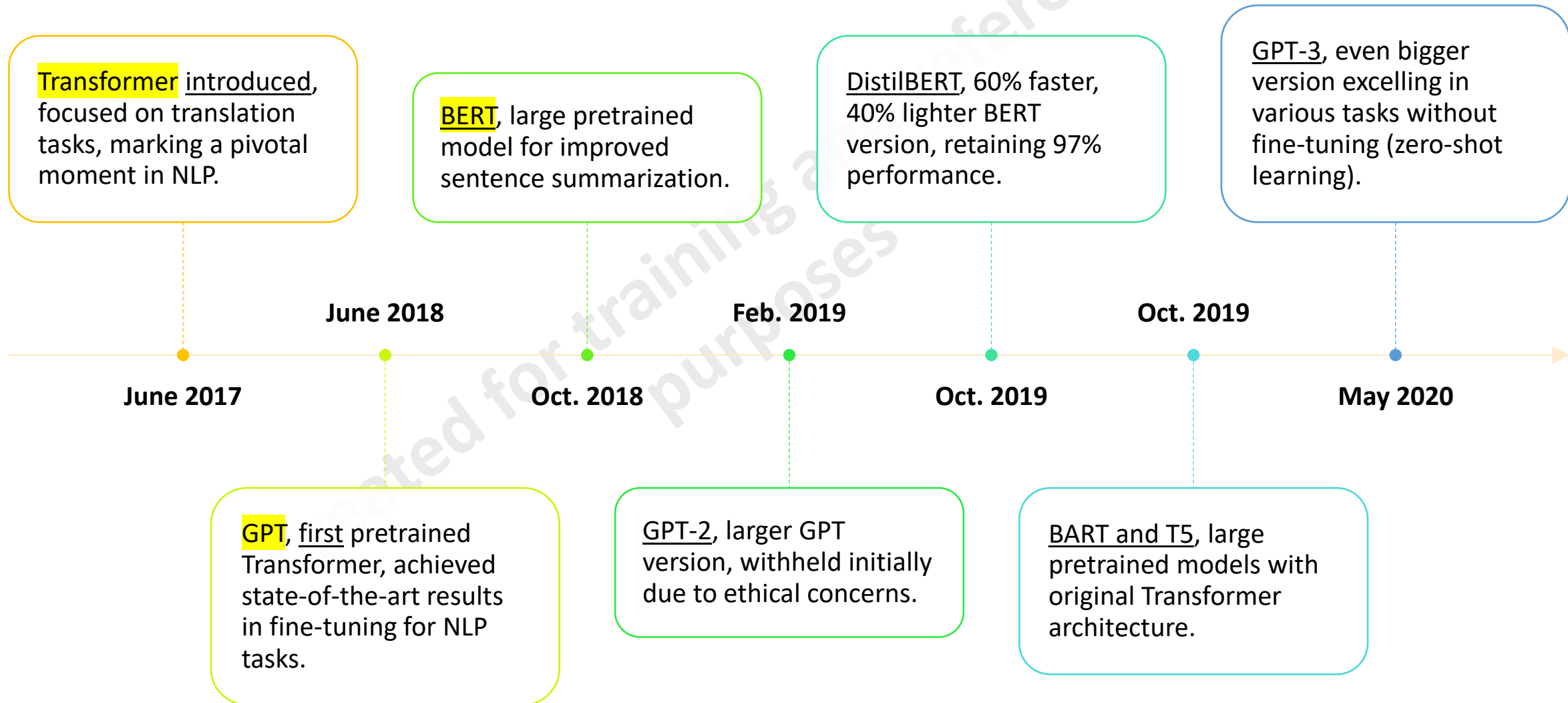
An introduction

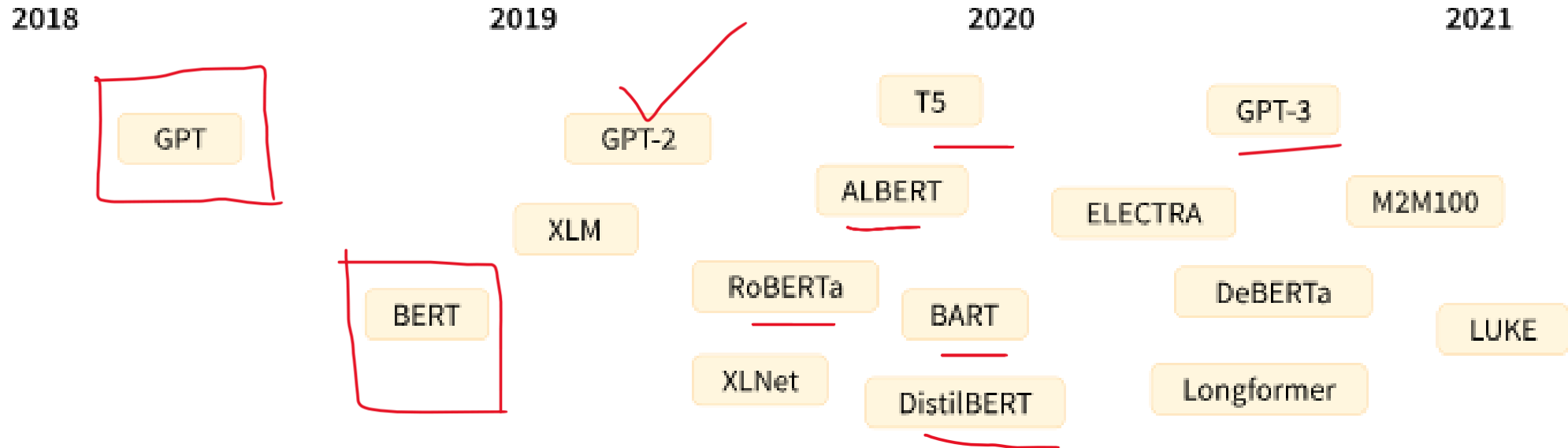
What is it?

- Transformer models represent a type of neural network architecture that has proven to be highly effective in various natural language processing (NLP) tasks and beyond.
- Introduced by Vaswani et al. in the paper "Attention is All You Need" in 2017, the Transformer architecture has become a cornerstone in the field of deep learning



Transformer history





Popular transformer models

5

Reasons to Learn About Transformer Models

Breakthrough in NLP

- represent a groundbreaking advancement in natural language processing (NLP) tasks

State-of-the-Art Performance

- demonstrated state-of-the-art performance across a wide range of NLP tasks, including language translation, text summarization, sentiment analysis, and question answering.

Versatility Across Domains

- Transformer architectures are not limited to NLP but have also been successfully applied to other domains such as computer vision and audio/speech processing.

Reasons to Learn About Transformer Models

Pretrained Models and Transfer Learning

- Transformer models are often **pretrained** on large-scale datasets, capturing rich linguistic representations that can be **fine-tuned** for specific downstream tasks with minimal data.

Scalability and Efficiency

- Transformers offer scalability and efficiency advantages compared to traditional recurrent neural networks (RNNs) and convolutional neural networks (CNNs) by allowing **parallel processing of input sequences**.

Career Opportunities:

- Proficiency in transformer models is highly valued in industry roles related to NLP, including data scientists, machine learning engineers, and NLP researchers.

Applications of transformer networks

translation between different languages.

Developing chatbots that are more interactive and practical.

Condensing lengthy documents into concise summaries.

Expanding a short prompt into a comprehensive document.

Generating chemical structures for drugs based on specific instructions.

Producing images based on textual descriptions.

Crafting descriptive captions for images.

Offering suggestions for code completion based on existing code snippets.



Transformer model implementations



Google introduced Bidirectional Encoder Representations from Transformers (BERT), one of the pioneering **LLMs** based on transformers.



OpenAI developed the GPT series, including iterations like GPT-2, GPT-3, GPT-3.5, GPT-4, and ChatGPT.



Meta's Llama achieves comparable performance despite its smaller size, being ten times smaller than similar models.



Google's Pathways Language Model demonstrates versatility by excelling in tasks across multiple domains, including text, images, and robotic controls.



OpenAI's Dall-E generates images based on textual descriptions.



The University of Florida and Nvidia's GatorTron analyze unstructured data from medical records.

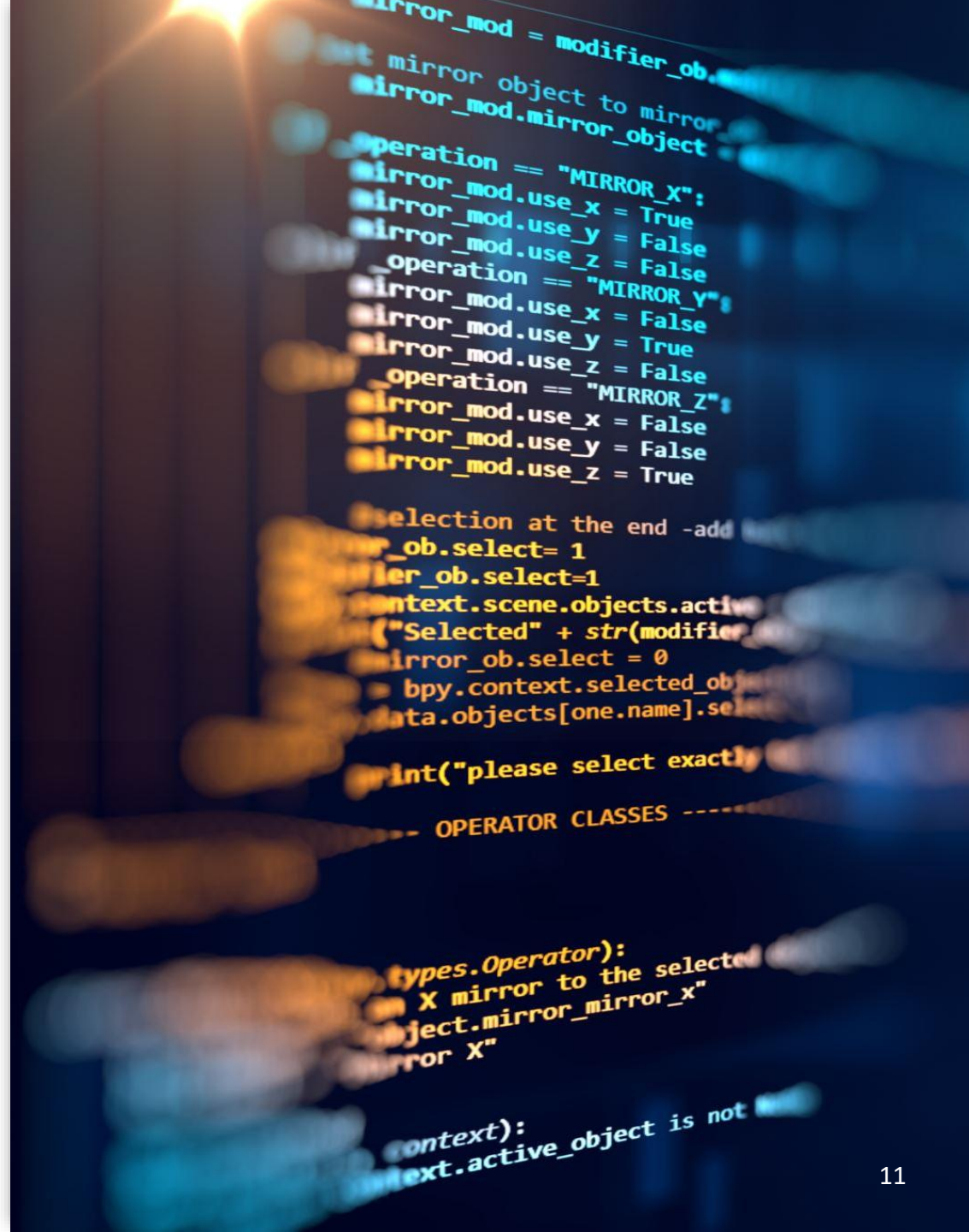
Reasons we need Transformer Models

- **Handling Long-Term Dependencies:**
Example: Language Translation
- **Shortcoming:** Earlier models like RNNs and LSTMs struggle to capture long-term dependencies in sequential data due to the vanishing gradient problem.
- **Solution:** Transformer models address this limitation by employing self-attention mechanisms, allowing them to capture relationships between distant tokens in a sequence effectively.



Reasons we need Transformer Models

- **Parallel Processing:** Example: Sentiment Analysis
- **Shortcoming:** Sequential models like RNNs process text tokens one at a time, resulting in slower inference times, especially for longer texts.
- **Solution:** Transformer-based sentiment analysis models can process entire sequences in parallel using self-attention, leading to faster inference and more efficient utilization of computational resources.

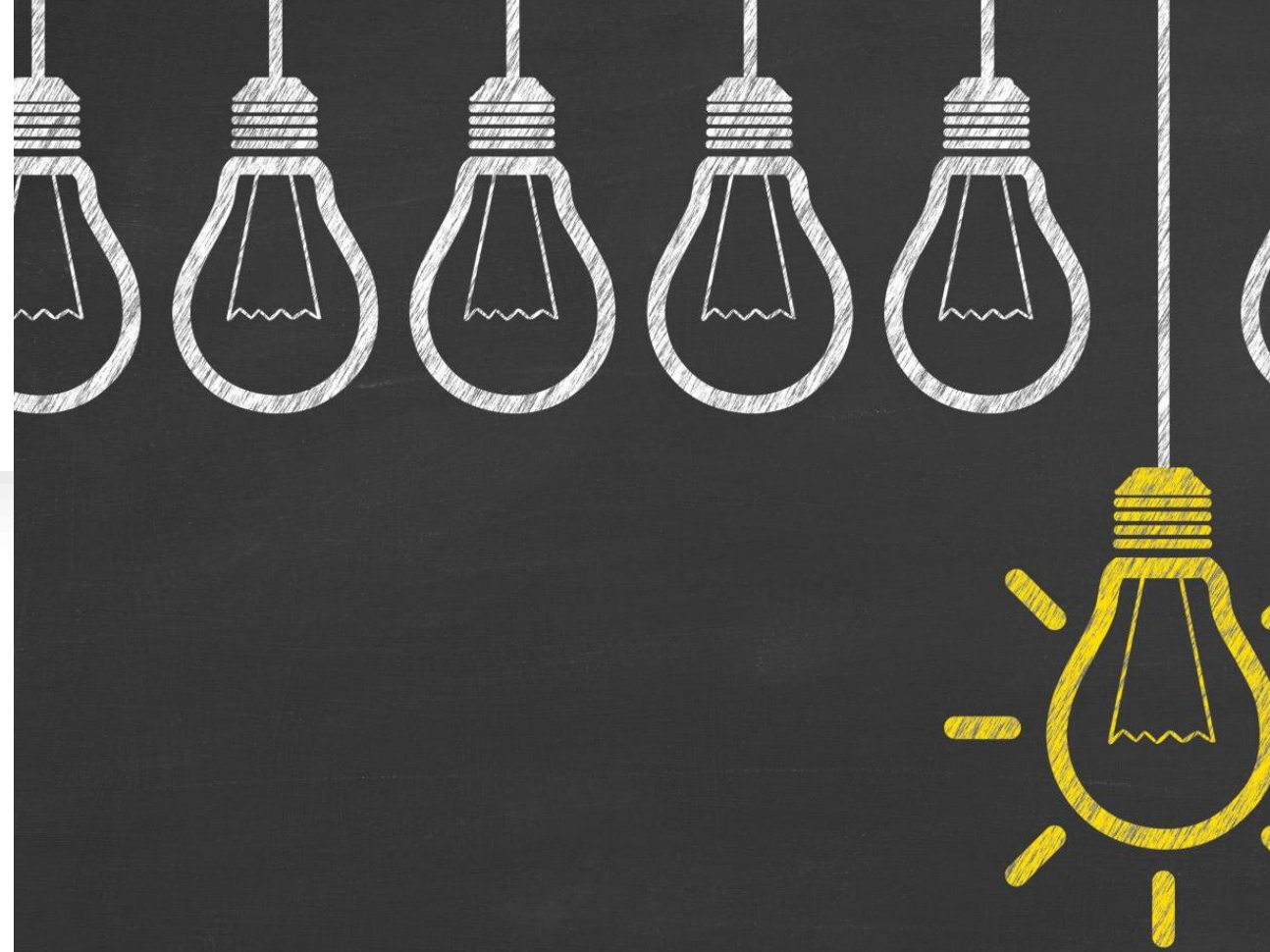


Reasons we need Transformer Models

- **Global Context Understanding:**
- Example: Document Summarization
- **Shortcoming:** RNN-based summarization models may struggle to capture the global context of a document, leading to incomplete or inaccurate summaries.
- **Solution:** Transformers consider all tokens in the document simultaneously, enabling them to capture the global context effectively and generate more informative and coherent summaries.

Reasons we need Transformer Models

- **Transfer Learning:**
- Example: Named Entity Recognition (NER)
- **Shortcoming:** Traditional NER models require substantial retraining or fine-tuning on domain-specific datasets, limiting their applicability to new domains.
- **Solution:** Pretrained transformer-based models like BERT can leverage knowledge learned from large-scale pretraining on diverse text corpora, enabling rapid adaptation to new NER tasks with minimal additional training.



Reasons we need Transformer Models

- **Contextual Understanding:**
- Example: Question Answering
- **Shortcoming:** Bag-of-words representations fail to capture the contextual relationships between words in a question and answer.
- **Solution:** Transformer-based QA models, such as BERT, understand the context of the question and passage simultaneously, enabling more accurate and contextually relevant answers.

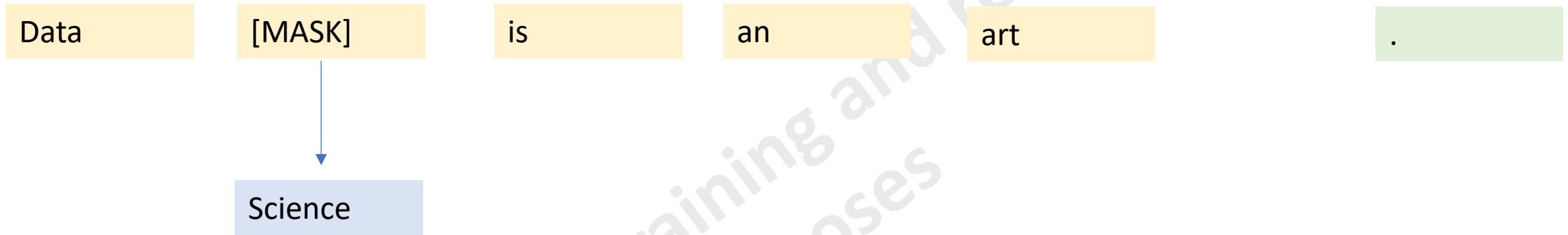


Example use case - next word in a sentence



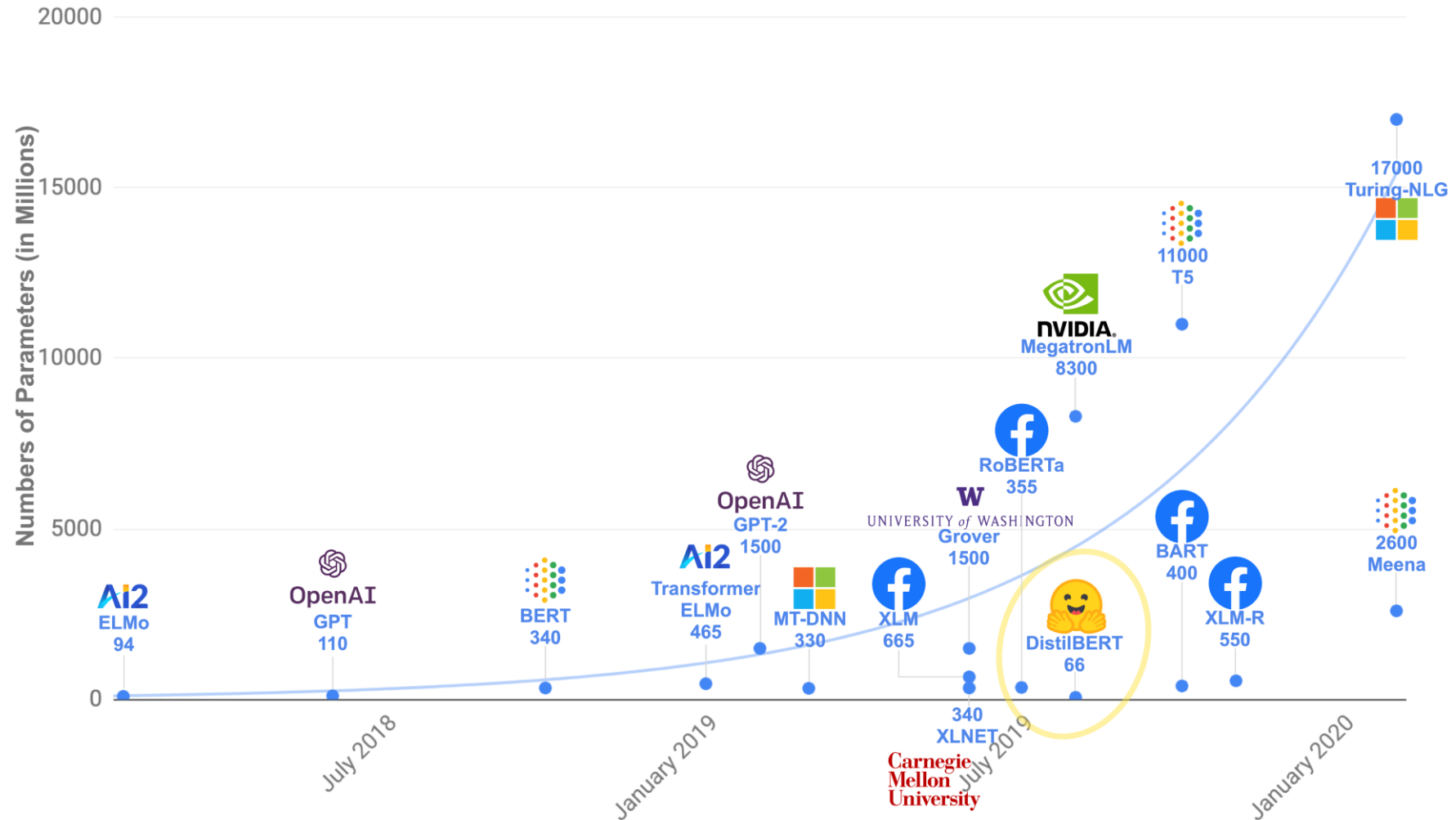
causal language modeling because the output depends on the past and present inputs, but not the future ones.

Another example



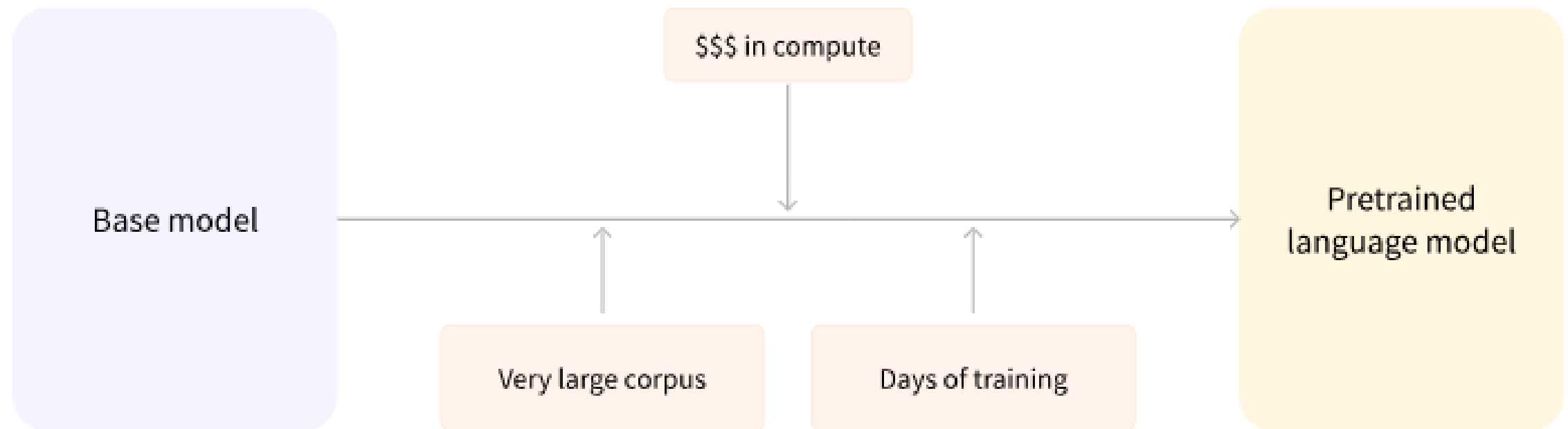
masked language modeling, in which the model predicts a masked word in the sentence.

Transformers are big models



Pretraining

- is the act of training a model from scratch:
 - the weights are randomly initialized, and the training starts without any prior knowledge.





What is Fine-Tuning

- **Definition:** Fine-tuning is a process in machine learning where a pre-trained model is further trained on a specific task or domain with a smaller, task-specific dataset.
- This helps the model adapt its learned representations to the nuances of the new task.

Examples of Fine-Tuning

- **Natural Language Processing (NLP):** Pre-trained language models (e.g., BERT, GPT) are fine-tuned on specific NLP tasks like sentiment analysis, named entity recognition, or machine translation.
- **Computer Vision:** Convolutional Neural Networks (CNNs) pre-trained on ImageNet are fine-tuned for specific image classification tasks in healthcare or autonomous vehicles.

Why fine tune?



**Knowledge
Transfer:**



**Dataset
Similarities:**



**Reduced Data
Requirements:**



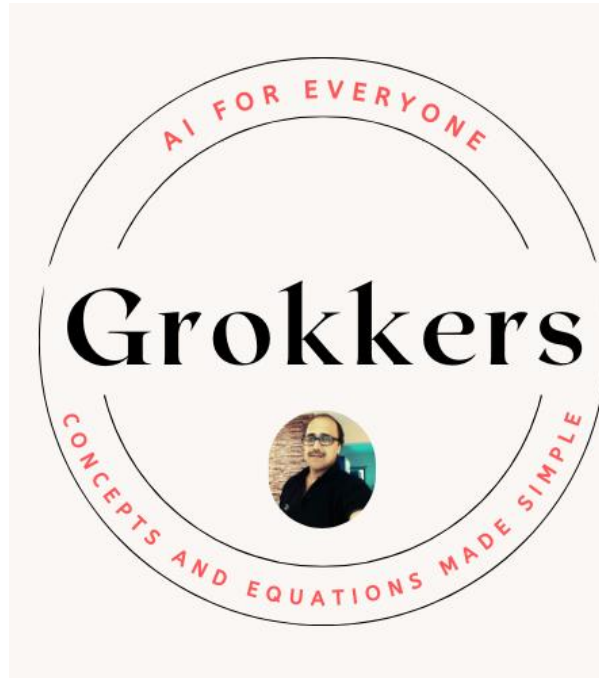
**Time and Resource
Efficiency:**



**Domain
Adaptation:**



**Effective Transfer
Learning:**



Thank you!